

10Pearls Data Science Internship Project Report

Pearls AQI Predictor:

**A Feature–Training–Inference Pipeline for
Three-Day-Ahead Air Quality Forecasting in Lahore**

Continuous US EPA AQI computed from PM2.5 | Horizons 24 / 48 / 72 hours



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1. Executive Summary

I built a three-day-ahead Air Quality Index forecast for Lahore as my 10Pearls Data Science Internship project. Pollution concentrations come from the OpenWeather Air Pollution API. Weather comes from Open-Meteo. I store engineered features and trained models in Hopsworks, optionally wrap inference in FastAPI, and show the forecast on a Streamlit dashboard at <https://lahore-aqi-predictor.streamlit.app>.

The target I predict is a **continuous US EPA AQI from PM2.5**, not OpenWeather's 1–5 category field. Most models learn the change in AQI over 24, 48 or 72 hours and I add that change back to the current AQI before scoring. I used RMSE, MAE and R^2 on the reconstructed absolute AQI. A model only goes into the Hopsworks registry if it beats a persistence baseline (tomorrow's AQI equals today's) on **all three** metrics.

My first full training notebook read feature group **aqi_features v3** (46,100 rows, 42 input features, December 2020 to August 2026). I used a season-aligned split: train to 1 June 2024 (27,947 rows), validation to 1 June 2025 (8,075 rows), and test from then on (10,078 rows). Optuna had 15 trials per tabular model. **Persistence was best at every horizon**. No learned model beat it on RMSE, MAE and R^2 together, so nothing was registered from that run.

I later restricted training to **4 April 2025** onwards after noticing a break in OpenWeather's AQI series (mean hourly |AQI change| drops from about 46 to about 4.5). On that shorter window the production models did beat persistence (RMSE **28.88 / 30.54 / 31.65** at 24 / 48 / 72 hours). I also tried classifying OpenWeather's 1–5 category instead of continuous AQI; that path did not beat “today's class” in a stable way, so I did not adopt it. **The production choice is continuous EPA AQI regression on the post-4 April 2025 window**. SHAP for those registered models is in Section 20.1. Section 17 is the first notebook; Section 20 is the later regression window; Section 21 is classification; Section 22 is the final choice.

2. Introduction

The project is a serverless Feature / Training / Inference pipeline. The flow I implemented is:

- OpenWeather (pollution) and Open-Meteo (weather) feed the feature pipeline.
- The feature pipeline writes to the Hopsworks Feature Store.
- The training pipeline writes to the Hopsworks Model Registry.
- FastAPI can serve a prediction; it is optional.
- Streamlit reads Hopsworks directly and shows the forecast.

Default location is Lahore at **31.5497, 74.3436**. History starts on **27 November 2020**, the first day OpenWeather's free air-pollution archive is available. I forecast 24, 48 and 72 hours. Smog season in this project is October through January.

On **4 April 2025** the OpenWeather series changes character: mean hourly |AQI change| collapses from about 46 to about 4.5 and does not recover, with the same break in PM2.5. I now default training to that date. The first notebook did not apply that filter; it trained on the full v3 table described in Section 9.

3. Problem Statement

The brief is to forecast a **continuous AQI** for Lahore at 24, 48 and 72 hours. OpenWeather's native `main.aqi` field is a coarse 1–5 category. RMSE, MAE and R^2 need a numeric target with meaningful distances, so I compute US EPA AQI from PM2.5 instead and keep the 1–5 field only as a reference column.

Hourly AQI is strongly autocorrelated, so “AQI in N hours equals AQI now” is a serious baseline, not a toy. I only register a model if it is strictly better on RMSE, strictly better on MAE, and strictly better on R^2 at the same time.

4. Project Objectives

- Forecast Lahore AQI at **24, 48 and 72 hours**.
- Compare continuous regression (0–500 EPA AQI) with classification of OpenWeather's 1–5 category, and keep only the approach that beats a serious baseline.
- Use one pollution source (OpenWeather) and one weather source (Open-Meteo) for both history and the hourly path.
- Store engineered features in Hopsworks (`aqi_features`).
- Register `aqi_forecaster_{24,48,72}h` only when a model beats persistence on RMSE, MAE and R^2 .
- Serve forecasts through FastAPI when needed, and present them in Streamlit.
- Automate hourly feature updates and daily retraining with GitHub Actions.

5. Project Scope

- City: Lahore at the default coordinates. The pipeline is location-parameterised; I did not run a multi-city training experiment.
- AQI is computed from **PM2.5 only**, using the 2024 US EPA PM2.5 breakpoint table. A full six-pollutant EPA maximum is out of scope.
- I fetch OpenWeather's 4-day pollution forecast for possible later comparison, but I do not use it in training.
- The first training notebook covers Ridge, RandomForest, XGBoost, LightGBM, a top-3 tabular mean ensemble, LSTM, GRU, Prophet, and persistence.
- `/predict` serves tabular or ensemble payloads that consume one feature row. Prophet needs a date series; LSTM/GRU need a 24-hour window, so those are not served that way.
- Production feature-group version is **4**. The first notebook used version **3**.

6. Project Workflow

1. Define the problem: continuous EPA AQI from PM2.5 at 24/48/72 hours for Lahore.
2. Collect OpenWeather pollution from 27 November 2020 and Open-Meteo weather for the same hours; merge on UTC timestamp.
3. Write a local raw snapshot (`aqi_raw_merged.parquet`).
4. Validate required columns, duplicates, nulls and physical bounds before insert.

5. Explore the raw snapshot (not the engineered feature group).
6. Engineer features on a strict hourly grid: time encodings, log1p pollutants, lags, rolling windows, change rates, weather interactions, absolute and delta targets.
7. Write the feature group to Hopsworks (v3 in the first notebook; v4 in current config).
8. Train and evaluate RMSE, MAE and R^2 on absolute future AQI versus persistence.
9. Register only if the model beats persistence on all three metrics.
10. Serve through Streamlit (and optional FastAPI) reading Hopsworks.

7. Dataset Description

7.1 Raw merged snapshot

I explored the local raw merge of pollution and weather. After loading I added month and a smog-season flag for plots; those two columns are not stored in the parquet.

Item	Details from local raw snapshot
File	aqi_raw_merged.parquet
Rows	46419
Stored columns	16
Timestamp range	2020-11-27 00:00:00 to 2026-08-14 13:00:00
Duplicate timestamps	0
Missing values (all stored columns)	none
Target used in this project	aqi (US EPA AQI from PM2.5)
Reference column (not the target)	openweather_aqi_category

Table 2: Raw snapshot used for EDA figures and descriptive statistics.

Stored columns and data types, before I added the two EDA-only columns:

Column	dtype on snapshot
timestamp	datetime64[ns]
openweather_aqi_category	float64
co	float64
no	float64
no2	float64

o3	float64
so2	float64
pm2_5	float64
pm10	float64
nh3	float64
aqi	int64
temperature	float64
humidity	int64
wind_speed	float64
wind_deg	int64
pressure	float64

Table 3: Raw snapshot columns and dtypes.

7.2 Data sources

Pollution and AQI come from the OpenWeather Air Pollution API (current, forecast and historical). Weather comes from Open-Meteo's archive and forecast APIs. I set the start date to 27 November 2020 so both sources share the first day. The Open-Meteo hourly fields I keep are temperature, humidity, wind speed, wind direction and surface pressure. All timestamps are UTC before the merge.

7.3 Target variable

I compute `aqi` from PM2.5 using the 2024-revised US EPA PM2.5 breakpoint table:

PM2.5 low ($\mu\text{g}/\text{m}^3$)	PM2.5 high ($\mu\text{g}/\text{m}^3$)	AQI low	AQI high	Label in source comment
0.0	9.0	0	50	Good
9.1	35.4	51	100	Moderate
35.5	55.4	101	150	Unhealthy for Sensitive Groups
55.5	125.4	151	200	Unhealthy
125.5	225.4	201	300	Very Unhealthy

225.5	325.4	301	500	Hazardous
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Table 4: PM2.5 breakpoints used to compute AQI.

Concentration is truncated (not rounded) to one decimal place before lookup, matching EPA methodology. Values above 325.4 $\mu\text{g}/\text{m}^3$ are extrapolated from the last band rather than hard-capped at 500. Official EPA AQI is the maximum across six criteria pollutants; I use **PM2.5 only**.

7.4 Feature-group table used in training

The first training notebook read the following from Hopsworks:

Item	Value
Feature group	aqi_features v3
Shape	(46100, 52)
Range	2020-12-05 18:00:00 -> 2026-08-08 06:00:00
Delta / target columns confirmed	aqi_delta_24h, aqi_delta_48h, aqi_delta_72h, aqi_target_24h, aqi_target_48h, aqi_target_72h
n_features after dropping timestamp, hour, month, openweather_aqi_category and all six target columns	42
Train rows	27947
Validation rows	8075
Test rows	10078
Train window printed	2020-12-05->2024-06-01
Validation window printed	2024-06-01->2025-06-01
Test window printed	2025-06-01->2026-08-08

Table 5: Feature group and split from the first training notebook.

Version 3 has delta targets but was built on a gappy frame, so about **13%** of lags and targets spanned the wrong number of hours. Version 4 is the same features on a strict hourly grid. The first notebook used v3; production now defaults to v4.

8. Data Understanding

On the raw snapshot I used for EDA:

Statistic	Value
aqi mean	247.7509855877981
aqi median	177.0
aqi standard deviation	194.00734285333604
aqi skewness	1.907488520950293
aqi kurtosis	3.1202724556139034
IQR outlier threshold (Q3 + 1.5 IQR)	479.0
Count of aqi values above that threshold	5837
ADF statistic	-11.764392029744617
ADF p-value	1.1280420946421232e-21

Table 6: AQI descriptive statistics and ADF test on the raw snapshot.

Yearly mean AQI after resampling to year-end:

Year-end timestamp	Mean aqi
2020-12-31	438.785818
2021-12-31	263.904701
2022-12-31	277.071900
2023-12-31	280.052722
2024-12-31	251.599010
2025-12-31	202.854102
2026-12-31	166.540349

Table 7: Yearly mean AQI on the raw snapshot. 2020 starts on 27 November; 2026 ends on 14 August.

I treat October–January as smog season. On the raw snapshot:

is_smog_season	count	mean	std	min	25%	50%	75%	max
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0 (Normal)	32406. 0	182.31540 5	126.14557 2	0.0	112. 0	159. 0	193. 0	1000. 0
1 (Smog Oct– Jan)	14013. 0	399.07514 5	234.69854 1	52. 0	207. 0	310. 0	553. 0	1000. 0

Table 8: *AQI grouped by smog-season flag on the raw snapshot.*

Pearson correlation of numeric columns with AQI (month excluded from the heatmap):

Variable	Correlation with aqi
aqi	1.000000
pm2_5	0.986036
pm10	0.958184
co	0.793539
no	0.560187
no2	0.546778
is_smog_season	0.512918
pressure	0.511338
openweather_aqi_category	0.477453
so2	0.367937
nh3	0.361643
humidity	0.284877
wind_deg	0.092419
wind_speed	-0.240057
o3	-0.266326
temperature	-0.545787

Table 9: *Correlation with aqi on the raw snapshot.*

Pollutant skewness, sorted descending: no 3.92, nh3 3.01, so2 2.46, no2 2.27, co 1.99, pm2_5 1.41, pm10 1.29, o3 1.26. That right-skew is why I apply log1p to those eight pollutant columns before training.

9. Data Preprocessing

9.1 Validation of raw merged data

I validate the merge before writing to Hopsworks. Required columns are timestamp, aqi, pm2_5, pm10, co, no2, o3, so2, nh3, temperature, humidity, wind_speed, wind_deg and pressure. Duplicate timestamps are dropped (keep first). Rows with nulls in required columns are dropped. Out-of-range rows are dropped using these bounds:

Column	Accepted range in VALID_RANGES
aqi	0 to 1000
pm2_5	0 to 2000 $\mu\text{g}/\text{m}^3$
pm10	0 to 2000 $\mu\text{g}/\text{m}^3$
co	0 to 50000 $\mu\text{g}/\text{m}^3$
no2	0 to 2000 $\mu\text{g}/\text{m}^3$
o3	0 to 2000 $\mu\text{g}/\text{m}^3$
so2	0 to 2000 $\mu\text{g}/\text{m}^3$
nh3	0 to 2000 $\mu\text{g}/\text{m}^3$
temperature	-30 to 60 $^{\circ}\text{C}$
humidity	0 to 100 %
wind_speed	0 to 150 km/h
wind_deg	0 to 360 degrees
pressure	850 to 1100 hPa

Table 10: Validation ranges applied before insert.

The bounds are wide on purpose so genuine smog extremes are not treated as errors. One bad hour drops that row; it does not abort the whole batch unless I set that flag.

9.2 Hourly grid and short-gap interpolation

I reindex onto a strict hourly calendar before any lag or target shift. The raw history has about **3,700 missing hours** in 415 outages. On the gappy frame a 24-hour shift spanned 24 hours for only 87% of rows and up to 264 hours for the rest. Outages of at most six hours, end to end, I interpolate linearly. Longer gaps stay missing and drop later. Integer-typed columns are rounded immediately after the grid step.

9.3 Feature transforms

11. Time features: hour, day_of_week, month, is_weekend, hour_sin, hour_cos, month_sin, month_cos.
12. Smog flag: is_smog_season for October–January.
13. log1p of clipped-at-zero co, no, no2, o3, so2, pm2_5, pm10, nh3 (raw columns replaced in place).
14. AQI lags at 1, 3, 6, 24 and 168 hours.
15. Rolling AQI mean, std, min, max over 3, 6 and 24 hours.
16. Change rates: aqi_change_rate_1h and aqi_change_rate_24h.
17. Weather interactions: wind_speed $\times$ pm2_5 and humidity $\times$ pm2_5.
18. Targets: aqi_target_{24,48,72}h and aqi_delta_{24,48,72}h.
19. Training mode: drop any row with a missing value. Inference mode: drop missing values only on feature columns so the latest hours (unknown future targets) remain.

9.4 Scaling

Ridge is a pipeline of StandardScaler plus Ridge. LSTM and GRU fit StandardScaler on training features and then transform train, validation and test. RandomForest, XGBoost and LightGBM are not scaled. Prophet uses timestamp and AQI only.

9.5 Train / validation / test split

I used a chronological split that snaps validation and test starts to 1 June. The run printed: train 2020-12-05→2024-06-01 | val 2024-06-01→2025-06-01 | test 2025-06-01→2026-08-08, with **27,947 / 8,075 / 10,078** rows. Optuna used TimeSeriesSplit with 5 folds on the training fold. Random state 42. 15 trials.

9.6 What I did not do in the first training run

That notebook did not start training on 4 April 2025. It also did not search delta shrinkage (that search exists in the production training loop). The raw snapshot had no duplicate timestamps, so duplicate removal was not needed there.

10. Exploratory Data Analysis

I plotted the raw snapshot using the same recipes as my EDA notebook (univariate AQI, pollutant histograms, AQI over time, AQI versus weather, correlation heatmap, ACF/PACF, seasonal decomposition, ADF, and smog versus normal). The notebook in git has no saved plot outputs, so I regenerated the figures below from the parquet.

10.1 AQI distribution

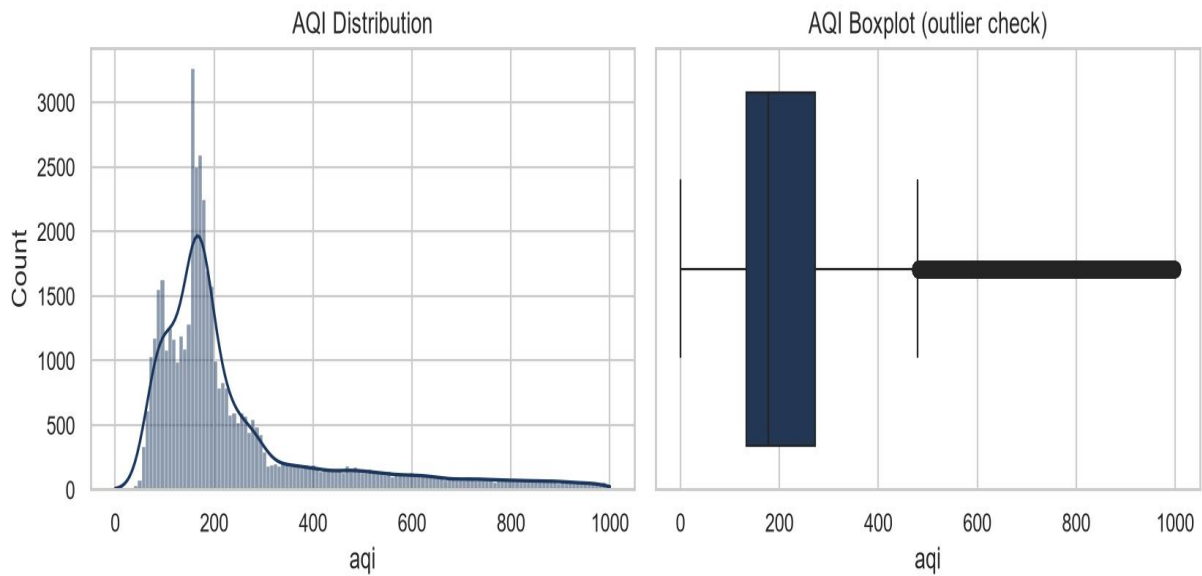


Figure 1: Distribution of *aqi* (histogram with KDE) and *aqi* boxplot on the raw snapshot.

AQI is right-skewed (**skewness 1.91**). The mean (247.8) sits well above the median (177). The boxplot has a long upper tail: **5,837** hours sit above the IQR fence of 479. The maximum in both smog and normal groups is 1000.

10.2 Pollutant distributions

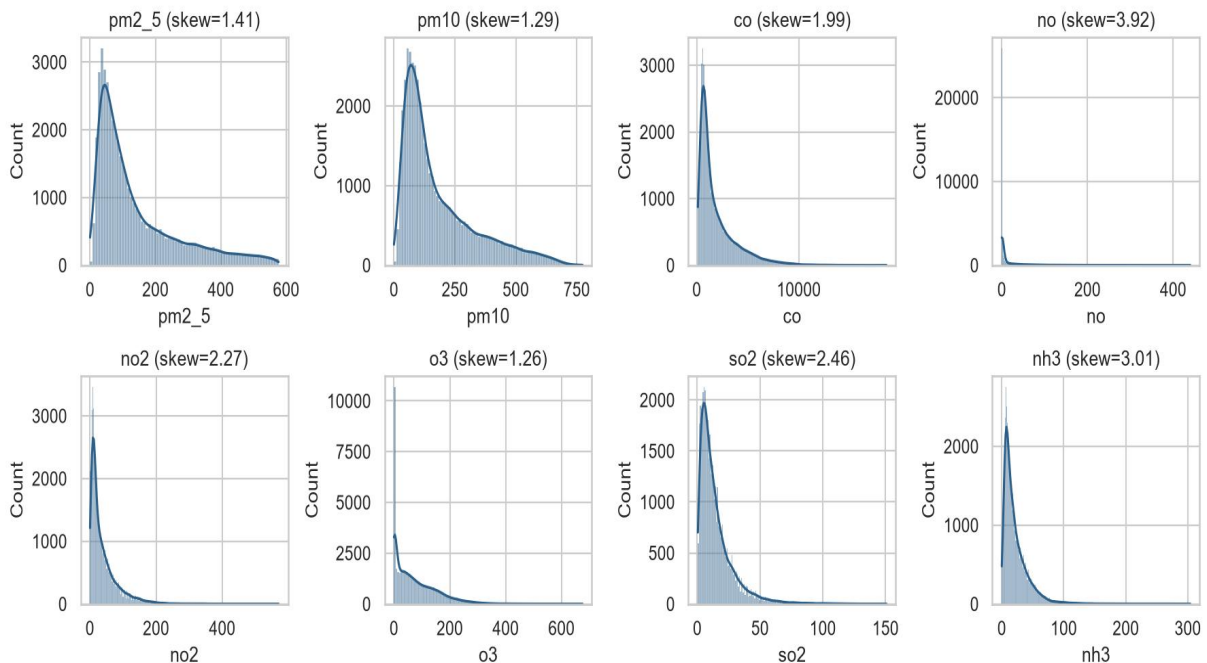


Figure 2: Histograms of *pm2_5*, *pm10*, *co*, *no*, *no2*, *o3*, *so2* and *nh3* on the raw snapshot.

All eight pollutant series are right-skewed. Ranked skewness is in Section 8. That is why I log-transform them before modelling.

10.3 AQI over time

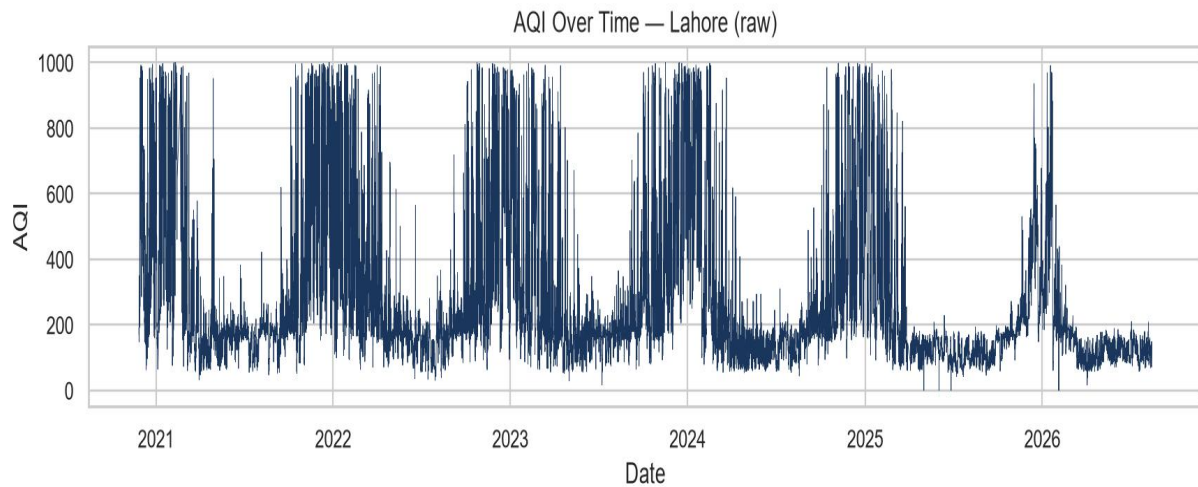


Figure 3: *aqi over time for Lahore on the raw snapshot (2020-11-27 to 2026-08-14).*

Winter peaks repeat every year, and the typical level drifts down later — yearly means in Table 7 fall from 438.8 in incomplete 2020 to 166.5 in incomplete 2026. That multi-year downward trend is one reason I train on **deltas** rather than raw level.

10.4 AQI versus weather

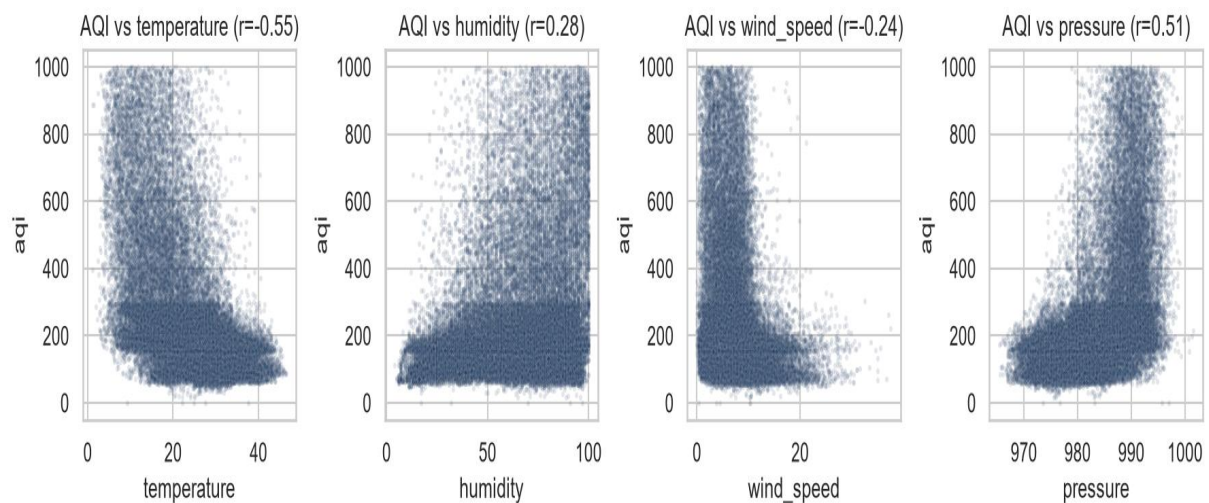


Figure 4: *Scatter of aqi versus temperature, humidity, wind_speed and pressure.*

Temperature correlates at -0.55 , pressure at $+0.51$, humidity at $+0.28$ and wind speed at -0.24 (Table 9). Cooler, higher-pressure hours tend to sit with worse AQI in this city. I treat those as associations, not a causal story, though wind dispersing pollution and humidity affecting secondary particles are the usual physical reading.

10.5 Correlation heatmap

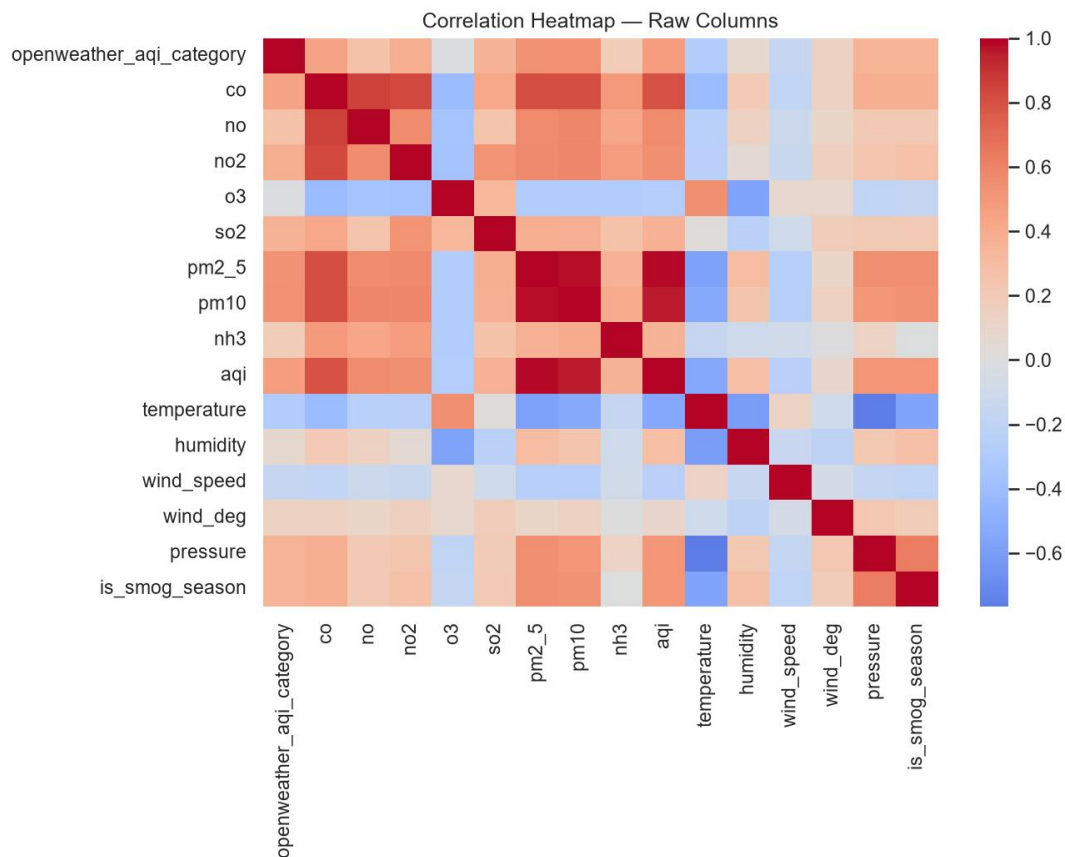


Figure 5: Correlation heatmap of numeric raw columns (month excluded).

AQI is almost collinear with **pm2_5 (0.986)** and very close to pm10 (0.958), which is expected because I compute AQI from PM2.5. OpenWeather's 1–5 category only correlates at 0.48, which is why I did not use it as the target.

10.6 Autocorrelation

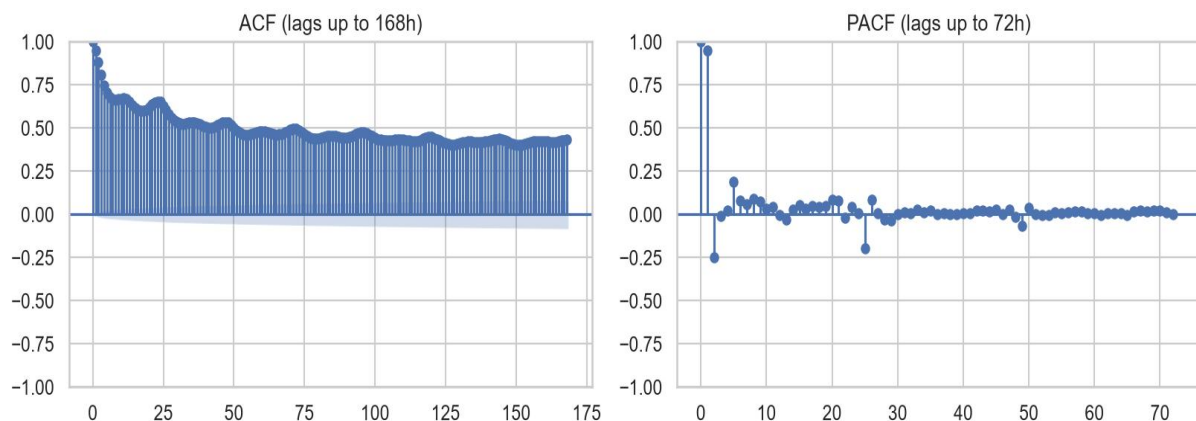


Figure 6: ACF of aqi up to 168 hourly lags and PACF of aqi up to 72 hourly lags.

Short-lag autocorrelation is strong, and the ACF still has structure at daily (24h) and weekly (168h) ranges. That is why I kept lags $\{1, 3, 6, 24, 168\}$ and rolling windows $\{3, 6, 24\}$.

10.7 Seasonal decomposition

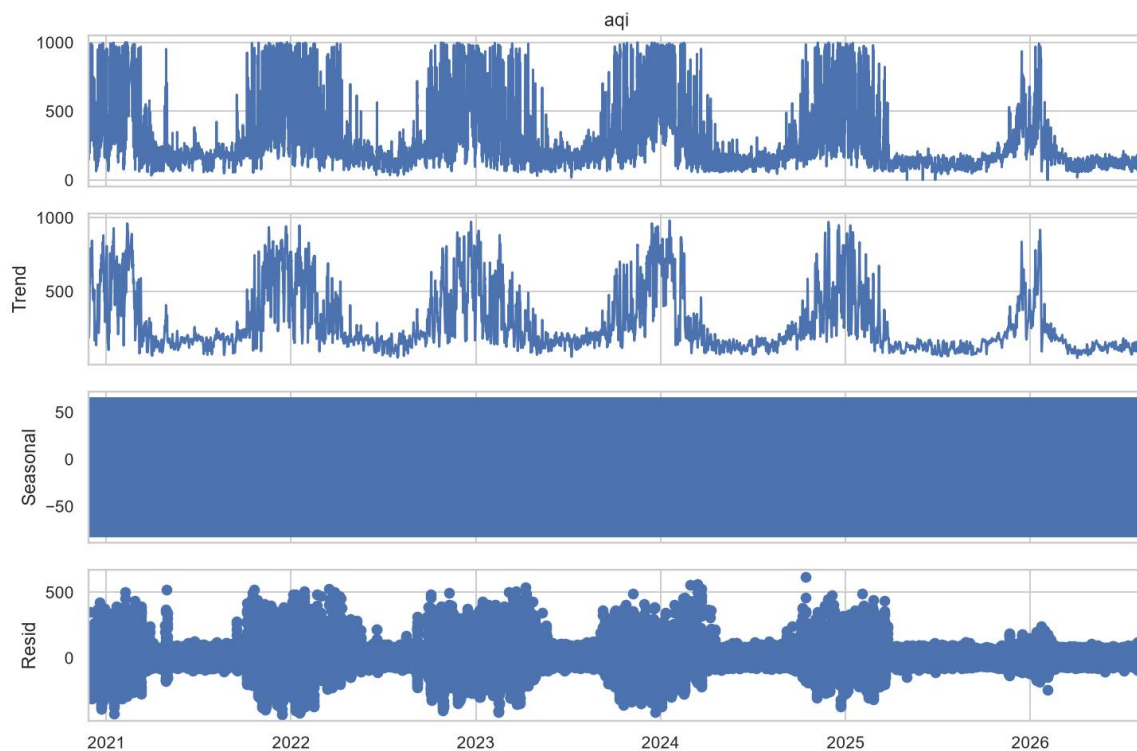


Figure 7: Additive seasonal_decompose of hourly aqi with period=24.

An additive 24-hour decomposition shows a slow trend, a repeating daily seasonal piece, and a residual. I interpolated to hourly frequency before decompose so the period is a true 24 hours.

10.8 Smog season versus normal season

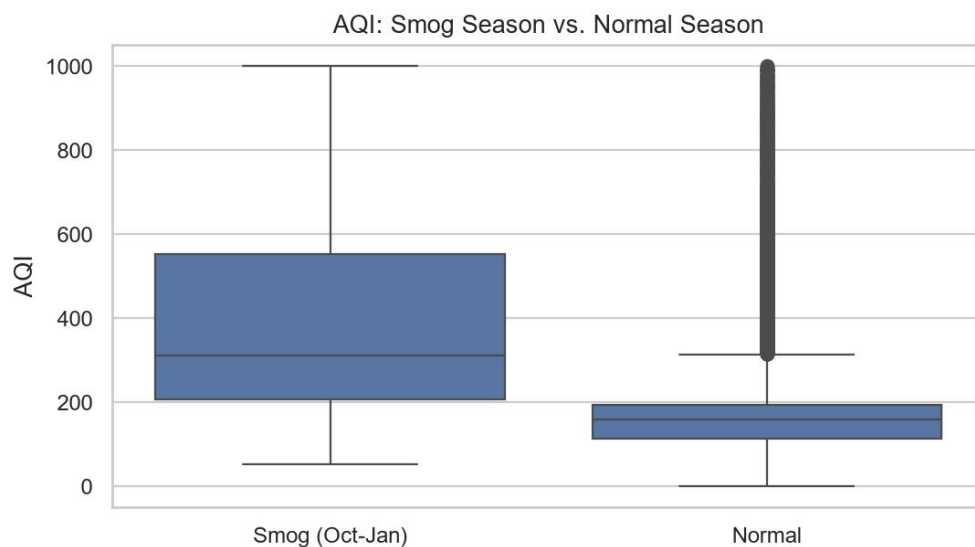


Figure 8: Boxplot of aqi for Normal versus Smog (Oct-Jan) on the raw snapshot.

Smog-season hours are much worse: mean **399** (n = 14,013) versus **182** (n = 32,406). I keep ``is_smog_season`` both as a model feature and as a way to slice the data.

11. Feature and Variable Analysis

The first notebook used **42 input columns**, after dropping timestamp, hour, month, the OpenWeather 1–5 field, and all six target columns from the 52-column feature table. The 42 names are:

- Pollutants after log1p (replaced in place): co, no, no2, o3, so2, pm2_5, pm10, nh3.
- Current aqi (untransformed).
- Weather: temperature, humidity, wind_speed, wind_deg, pressure.
- Time encodings kept as features: day_of_week, is_weekend, hour_sin, hour_cos, month_sin, month_cos, is_smog_season.
- AQI lags: aqi_lag_1h, aqi_lag_3h, aqi_lag_6h, aqi_lag_24h, aqi_lag_168h.
- Rolling aqi: mean/std/min/max for windows 3h, 6h and 24h (12 columns).
- Change rates: aqi_change_rate_1h, aqi_change_rate_24h.
- Interactions: wind_pollutant_interaction, humidity_pollutant_interaction.

That list has 42 names. The notebook printed ``n_features: 42`` but did not dump the names as a separate list; I reconstructed them from the feature builder plus the drop list.

Tabular models, LSTM and GRU train on ``aqi_delta_{h}h``. Absolute metrics use ``aqi_target_{h}h``. Persistence predicts the current ``aqi`` column. Prophet is fit on timestamp and AQI and predicts at test time plus h hours.

12. Methodology

All families except Prophet predict the AQI change over the horizon. I then add that delta to current AQI. Metrics are RMSE, MAE and R^2 on the absolute future AQI. Persistence is scored the same way against current AQI.

Registration rule: RMSE must be **lower** than persistence, MAE **lower**, and R^2 **higher**. Among eligible models I sort by RMSE, then MAE, then $-R^2$. If none are eligible, I register nothing.

That first run logged into Hopsworks project <https://eu-west.cloud.hopsworks.ai:443/p/41103> and read ``aqi_features`` v3. TensorFlow saw a GPU; XGBoost used CUDA.

13. Approaches Considered and Used

13.1 Persistence baseline

No-change forecast: future AQI equals current AQI. No extra features and no training. I score it with RMSE, MAE and R^2 on the held-out test fold (Tables 11–13 in the per-model section, and Tables 21–23 in the full comparison). In the first notebook this baseline **won every horizon**, so nothing else was registered.

13.2 Optuna-tuned tabular regression on deltas

Ridge, RandomForest, XGBoost and LightGBM predict the horizon delta from the 42 features. I tuned each with Optuna (15 trials, time-series CV RMSE) and refit on all training rows. None beat persistence on all three metrics in that run.

13.3 Mean ensemble of top-3 tabular models

I averaged the reconstructed absolute predictions of the three tabular models with lowest test RMSE. That ensemble was not eligible under the persistence gate either.

13.4 Sequence models (LSTM and GRU)

LSTM and GRU see 24-hour windows of the 42 scaled features and predict the horizon delta. Architecture: Input (24, n_feat), LSTM/GRU 64 with return_sequences, Dropout 0.2, LSTM/GRU 32, Dropout 0.2, Dense 16 ReLU, Dense 1. Adam at 1e-3, MSE loss, 100 epochs, batch 128, early stopping on val_loss with patience 8. Both had RMSE above 219 and **negative R²** on every horizon. I did not register them.

13.5 Prophet on absolute AQI

Prophet models the AQI level directly. I fit it on train timestamp and AQI with daily and weekly seasonality on, yearly seasonality off. Future dates are test timestamp plus the horizon. RMSE was above **2150** on every horizon. Not registered.

13.6 Production training window

After the OpenWeather break on 4 April 2025 I started training from that date only. That is a different experiment from the first notebook. Results for that window are in Section 20, not mixed into Section 17.

13.7 Classification of OpenWeather 1–5

OpenWeather also publishes a 1–5 air-quality class. I ran that as a separate experiment: predict tomorrow's class from the same features, with persistence meaning "today's class." Models were ordinal logistic regression, Random Forest, XGBoost, LightGBM, a majority-vote ensemble, LSTM and GRU (5-way softmax). A model had to beat persistence on accuracy, macro F1, RMSE, MAE and R² together (the last three on the 1–5 labels). Results are in Section 21. I did not adopt this path.

14. Models Used

Numbers below are the full-precision test scores from the first training notebook. One-line prints in that notebook used fewer decimals; the tables keep the full comparison.

14.1 Persistence Baseline

No-change forecast, used as the registration gate. Input is current AQI only. No training.

Horizon	RMSE	MAE	R2
24	66.730405	36.111431	0.745204
48	84.796609	45.775352	0.588437
72	92.456124	51.518059	0.510480

Table 11: Persistence Baseline test metrics.

On that test window, predicting no change already explains a large share of variance at 24h (**R² 0.745**) and still more than half at 72h (R² 0.510). RMSE rises with horizon. Under the gate I wrote, this is the selected forecast from that notebook.

14.2 Ridge

Linear model of the AQI delta with L2 penalty, after StandardScaler. I searched alpha in [0.01, 100] (log-uniform, 15 trials) and then fit StandardScaler + Ridge on the training fold.

Horizon	RMSE	MAE	R2
24	87.584091	63.751444	0.561069
48	101.382767	73.917026	0.411688
72	104.353359	77.654450	0.376391

Table 12: Ridge test metrics.

Ridge RMSE and MAE are higher than persistence at every horizon, and R² is lower. I did not select it.

14.3 RandomForest

Bagged trees on the AQI delta. Optuna over n_estimators 100–250 (step 50), max_depth in {8, 12, 16}, min_samples_leaf 1–8, with max_samples=0.8, 15 trials.

Horizon	RMSE	MAE	R2
24	84.914023	59.164036	0.587424
48	118.050540	86.744712	0.202344
72	116.121902	85.821522	0.227804

Table 13: RandomForest test metrics.

Worse than persistence on RMSE, MAE and R² at every horizon. Not selected.

14.4 XGBoost

Gradient-boosted trees on the AQI delta. Optuna over learning_rate 0.01–0.3 (log-uniform), max_depth 3–9, n_estimators 100–400 (step 50), subsample and colsample_bytree 0.6–1.0. This run used hist + CUDA.

Horizon	RMSE	MAE	R2
24	74.364967	50.019281	0.683567
48	97.213262	69.199877	0.459083
72	111.990073	82.949948	0.281779

Table 14: XGBoost test metrics.

Among learned models, XGBoost had the lowest 24h and 48h RMSE, but both RMSE and MAE remain above persistence and R² remains below it. Closest at 24h (**74.36 vs 66.73**), still not eligible. Not selected.

14.5 LightGBM

Gradient-boosted trees on the AQI delta. Optuna over learning_rate 0.01–0.3, num_leaves 16–96, n_estimators 100–400 (step 50), subsample and colsample_bytree 0.6–1.0.

Horizon	RMSE	MAE	R2
24	76.456676	52.289149	0.665515
48	101.157242	72.808352	0.414302
72	112.211034	81.688058	0.278942

Table 15: LightGBM test metrics.

Close to XGBoost at 24h, still behind persistence on all three metrics. Not selected.

14.6 Ensemble_top3_tabular

Mean of the three best tabular absolute reconstructions by test RMSE. No extra fit; just an average of models already trained.

Horizon	RMSE	MAE	R2
24	77.085201	53.331851	0.659993
48	98.112376	70.283343	0.449031
72	107.619632	79.190920	0.336742

Table 16: *Ensemble_top3_tabular test metrics.*

The ensemble did not beat the best single tabular model at 24h or 48h and did not beat persistence. Not selected.

14.7 LSTM

Recurrent net on 24-hour windows of scaled features, predicting the delta. Training setup is in Section 13.4.

Horizon	RMSE	MAE	R2
24	232.032646	156.863391	-2.075345
48	234.142256	152.548810	-2.132563
72	235.968967	160.365461	-2.182948

Table 17: *LSTM test metrics.*

RMSE more than three times persistence at 24h; R² is negative, so the forecast is worse than predicting the test-set mean. Not selected.

14.8 GRU

Same sequence setup as LSTM, with GRU layers.

Horizon	RMSE	MAE	R2
24	219.023846	146.531186	-1.740176
48	246.401500	157.766790	-2.469181
72	239.031295	159.196560	-2.266098

Table 18: *GRU test metrics.*

Slightly better than LSTM at 24h, still far worse than persistence, with negative R². Not selected.

14.9 Prophet

Additive time-series model of absolute AQI. Inputs are train timestamp and train AQI only.

Horizon	RMSE	MAE	R2
24	2167.324472	1736.633538	-267.777676

48	2159.443832	1727.409530	-265.909192
72	2150.783604	1715.485984	-263.906508

Table 19: Prophet test metrics.

RMSE above 2150 AQI points on every horizon. Yearly seasonality was off. I did not dig further into fitted parameters after that. Not selected.

15. Model Training

Per horizon I computed persistence first, then trained Prophet, then tuned Ridge, RandomForest, XGBoost and LightGBM on training deltas, then trained LSTM and GRU, then built the top-3 tabular ensemble, then printed a table sorted by RMSE and applied the registration gate. If none passed, I registered nothing.

The notebook can write per-horizon comparison CSVs. The numbers in this report come from the saved cell output in that notebook, not from those CSVs.

16. Model Evaluation

On the regression path I used RMSE, MAE and R^2 on absolute future AQI (0–500). The classification path in Section 21 also used accuracy and macro F1 on the 1–5 labels.

The gate is: RMSE smaller than persistence, MAE smaller, R^2 larger. The printed outcome for each horizon was that **no model beat persistence on all three metrics**, so nothing was registered:

horizon	registered	winner
24	False	None
48	False	None
72	False	None

Table 20: Registry summary from the first training notebook.

17. Results

17.1 Complete comparison (24 hours)

Model	RMSE	MAE	R2
Persistence Baseline	66.730405	36.111431	0.745204
XGBoost	74.364967	50.019281	0.683567

LightGBM	76.456676	52.289149	0.665515
Ensemble_top3_tabular	77.085201	53.331851	0.659993
RandomForest	84.914023	59.164036	0.587424
Ridge	87.584091	63.751444	0.561069
GRU	219.023846	146.531186	-1.740176
LSTM	232.032646	156.863391	-2.075345
Prophet	2167.324472	1736.633538	-267.777676

Table 21: Test comparison for horizon 24h.

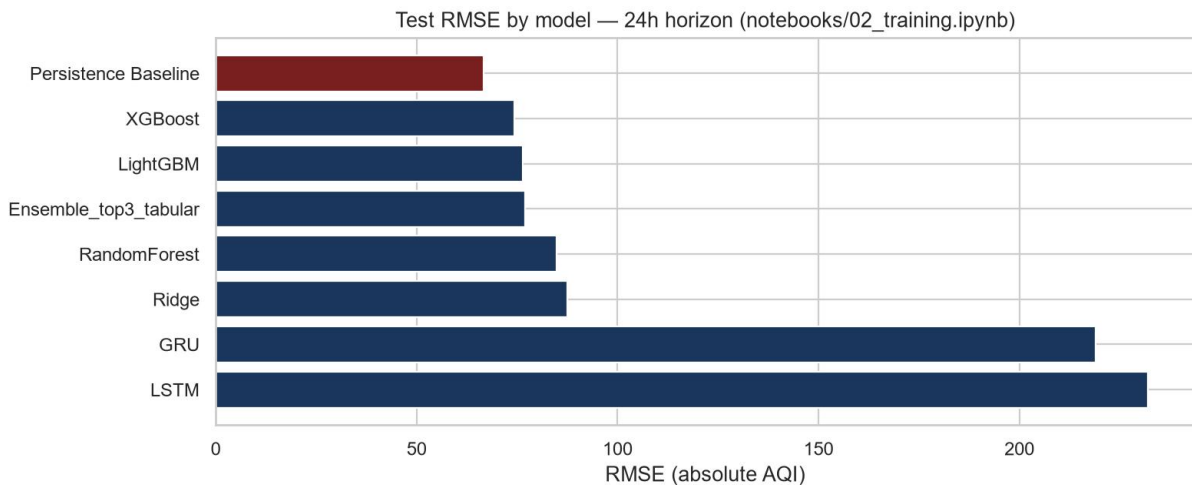


Figure 9: Test RMSE by model at 24h using the values in Table 21. Persistence is the top bar after sorting by RMSE. Prophet is omitted from this bar chart because RMSE 2167.324472 would flatten the other bars; Prophet remains in Table 21.

Every learned model has higher RMSE and higher MAE than persistence, and lower R^2 . XGBoost is the closest learned model (RMSE 74.36 versus 66.73).

17.2 Complete comparison (48 hours)

Model	RMSE	MAE	R2
Persistence Baseline	84.796609	45.775352	0.588437
XGBoost	97.213262	69.199877	0.459083
Ensemble_top3_tabular	98.112376	70.283343	0.449031
LightGBM	101.157242	72.808352	0.414302
Ridge	101.382767	73.917026	0.411688

RandomForest	118.050540	86.744712	0.202344
LSTM	234.142256	152.548810	-2.132563
GRU	246.401500	157.766790	-2.469181
Prophet	2159.443832	1727.409530	-265.909192

Table 22: Test comparison for horizon 48h.

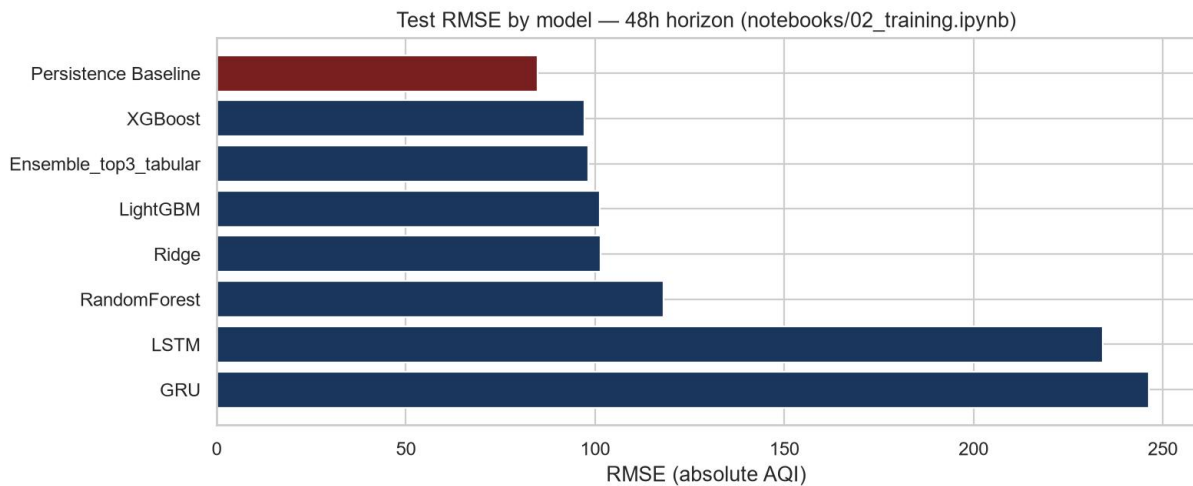


Figure 10: Test RMSE by model at 48h using the values in Table 22. Prophet omitted from bars for the same scale reason as Figure 9.

Persistence remains first (**RMSE 84.80**). XGBoost is again the closest learned model (97.21). LSTM, GRU and Prophet remain far worse.

17.3 Complete comparison (72 hours)

Model	RMSE	MAE	R2
Persistence Baseline	92.456124	51.518059	0.510480
Ridge	104.353359	77.654450	0.376391
Ensemble_top3_tabular	107.619632	79.190920	0.336742
XGBoost	111.990073	82.949948	0.281779
LightGBM	112.211034	81.688058	0.278942
RandomForest	116.121902	85.821522	0.227804
LSTM	235.968967	160.365461	-2.182948
GRU	239.031295	159.196560	-2.266098
Prophet	2150.783604	1715.485984	-263.906508

Table 23: Test comparison for horizon 72h.

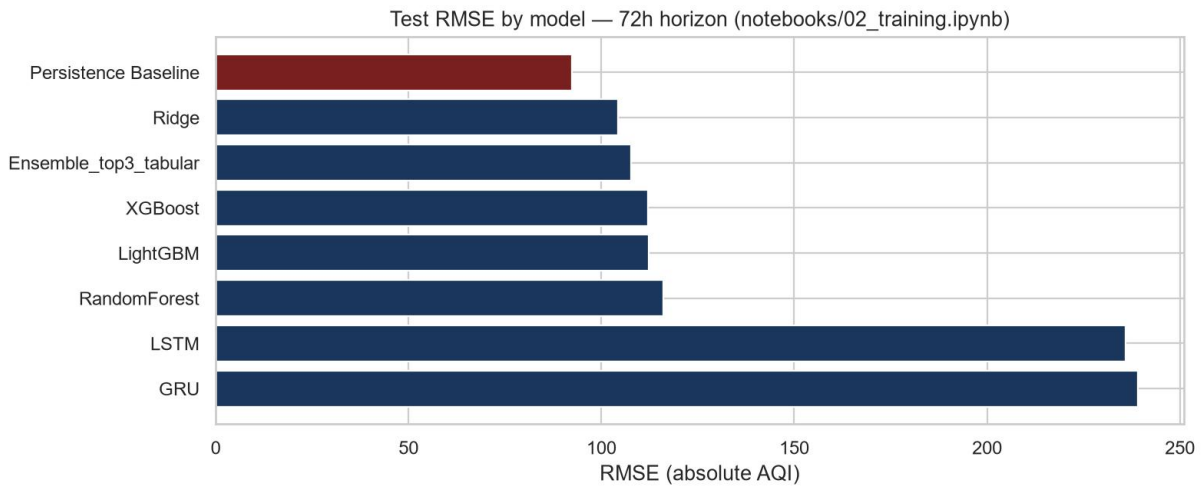


Figure 11: Test RMSE by model at 72h using the values in Table 23. Prophet omitted from bars for the same scale reason as Figure 9.

Persistence RMSE **92.46** is still lowest. Among learned models, Ridge has the lowest 72h RMSE (104.35), still above persistence.

18. Model Comparison

Table 24 puts persistence beside the best learned model by RMSE at each horizon. “Best learned” means lowest RMSE among non-persistence rows. It is not a registration winner.

Horizon	Persistence RMSE	Best learned model by RMSE	That model RMSE	Selected
24	66.730405	XGBoost	74.364967	No — persistence wins
48	84.796609	XGBoost	97.213262	No — persistence wins
72	92.456124	Ridge	104.353359	No — persistence wins

Table 24: Persistence versus lowest-RMSE learned model.

19. Best Model on the First Training Notebook

On the first notebook (full history, feature group v3) I selected **no learned model**. Persistence is the only forecast that satisfies the three-metric rule on that split. Exact

persistence numbers are in Table 11. Registered was False and winner was None for 24, 48 and 72 hours.

That is not my production choice. It is the result of that one experiment. The later regression window is in Section 20. Classification is in Section 21. I state the final decision in Section 22.

20. Production Training Window

This section is a later experiment. Training across the 4 April 2025 break is what made every model lose to persistence on the 2025–26 test period, so I now start training on that date. Selected models for that window:

Horizon	Selected model	RMSE	MAE	R ²
24 h (day 1)	LightGBM (L1)	28.88	22.63	0.064
48 h (day 2)	Random Forest	30.54	24.89	−0.055
72 h (day 3)	Ensemble (top-3 tabular)	31.65	26.07	−0.098

Table 25: Selected models on the post-4 April 2025 training window. Not from the first notebook.

Horizon	Persistence RMSE	Selected-model RMSE
24 h	34.84	28.88
48 h	40.94	30.54
72 h	40.30	31.65

Table 26: Persistence versus selected RMSE on the post-4 April 2025 window.

After mentor review I treated this window as the production choice, with registry names `aqi\_forecaster\_24h`, `aqi\_forecaster\_48h` and `aqi\_forecaster\_72h`. I do not replace Tables 21–23 with Tables 25–26. The two experiments differ in training start date and in feature-group version (v3 then, v4 now).

20.1 SHAP analysis of registered models

The first notebook has a SHAP block, but it only runs when a model is registered. That run registered nothing, so it produced no SHAP plots.

I therefore computed SHAP myself on the models currently cached from the registry, using TreeExplainer — the same method the training pipeline uses. Features come from the local raw parquet restricted to **4 April 2025** onwards. The chronological split fell back to 70/15/15 on that window: **6,956 / 1,491 / 1,491** rows (test from 29 May 2026 11:00 to 11 August 2026

13:00). I sampled 400 test rows for gradient-boosted trees and **150** for RandomForest (a 2,000-row forest run timed out).

The cached payloads are named **XGBoost** (24h), **RandomForest** (48h) and **Ensemble_top3_tabular** (72h members XGBoost_L1, RandomForest, LightGBM). Those names come from the saved models, not from Table 25. SHAP explains the predicted AQI delta (target_type delta, shrinkage 1.0).

20.1.1 24-hour registered model (XGBoost)

Rank	Feature	Mean SHAP
1	pressure	5.7132248878479
2	aqi_change_rate_24h	3.0580039024353027
3	month_cos	2.942716598510742
4	pm10	2.813547134399414
5	temperature	2.3712317943573
6	humidity_pollutant_interaction	2.0532147884368896
7	wind_deg	1.9747484922409058
8	day_of_week	1.7397871017456055
9	aqi_roll_min_24h	1.629075527191162
10	aqi_roll_min_6h	1.5423707962036133

Table 27: Top 10 mean |SHAP| features for the 24h XGBoost model (400 test rows).

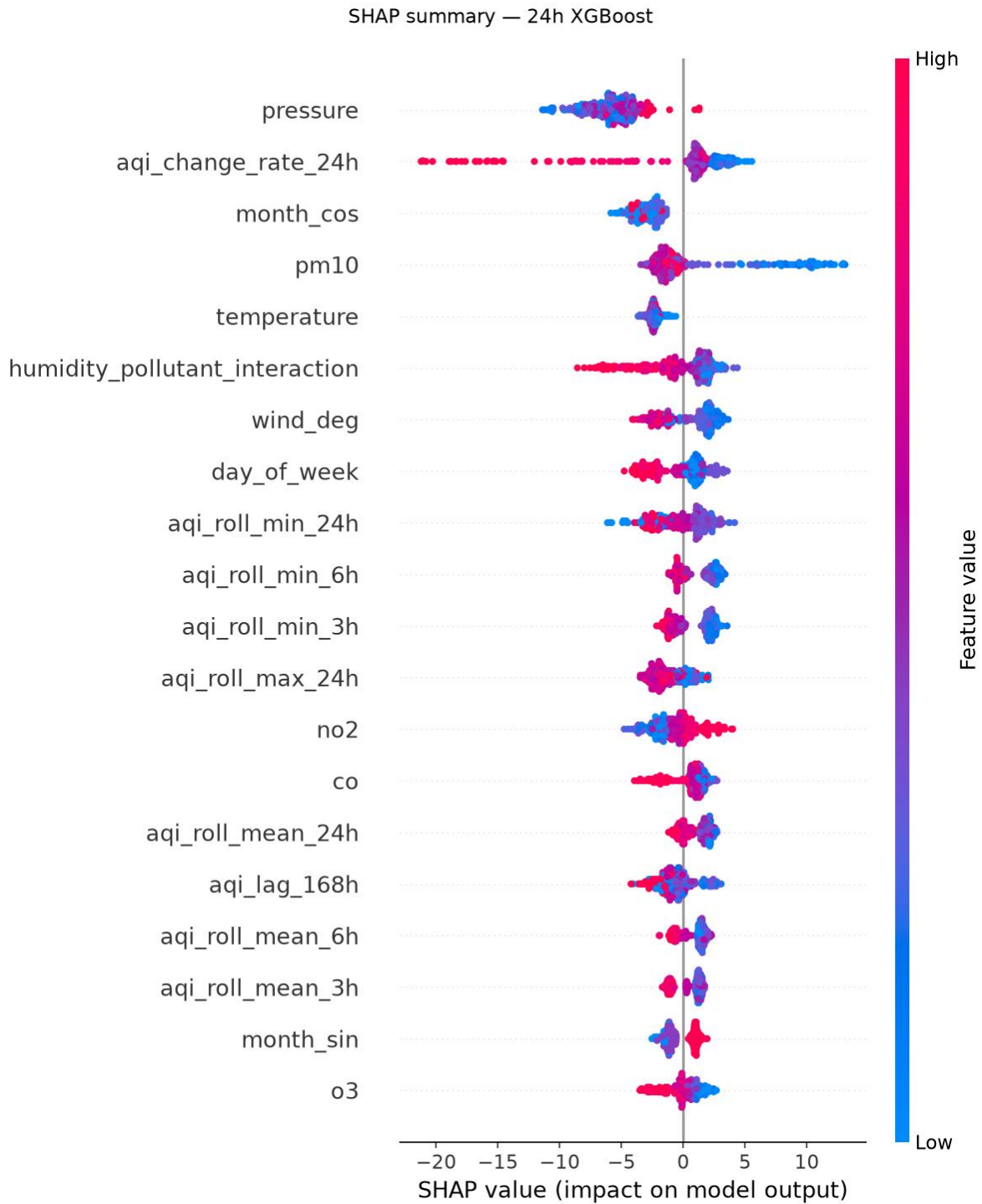


Figure 12: SHAP summary (beeswarm) for the 24h XGBoost model.

Each point is one sampled test hour. Colour is the feature value (high versus low). Horizontal position is that feature's contribution to the predicted delta. **pressure** has the largest mean $|\text{SHAP}|$ (5.71), then **aqi_change_rate_24h** (3.06) and **month_cos** (2.94).

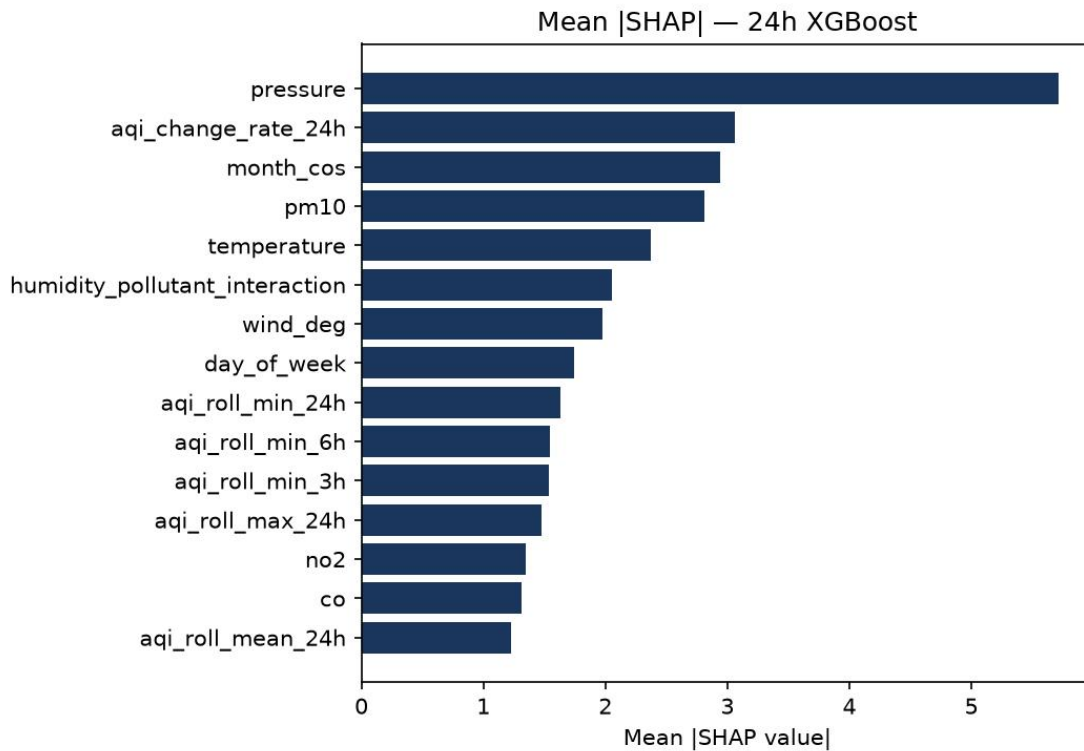


Figure 13: Mean |SHAP| bar chart for the 24h XGBoost model (top 15 features).

20.1.2 48-hour registered model (RandomForest)

Rank	Feature	Mean SHAP
1	pm2_5	8.678609856675047
2	pressure	5.349003206462411
3	month_cos	4.57826334528948
4	aqi_change_rate_24h	3.6110002145874653
5	aqi_lag_168h	3.0203817288733674
6	day_of_week	2.8400790645516274
7	month_sin	2.4771160689993983
8	temperature	2.2241804783036776
9	aqi_roll_max_24h	1.9557856727520853
10	nh3	1.48097618134355

Table 28: Top 10 mean |SHAP| features for the 48h RandomForest model (150 test rows).

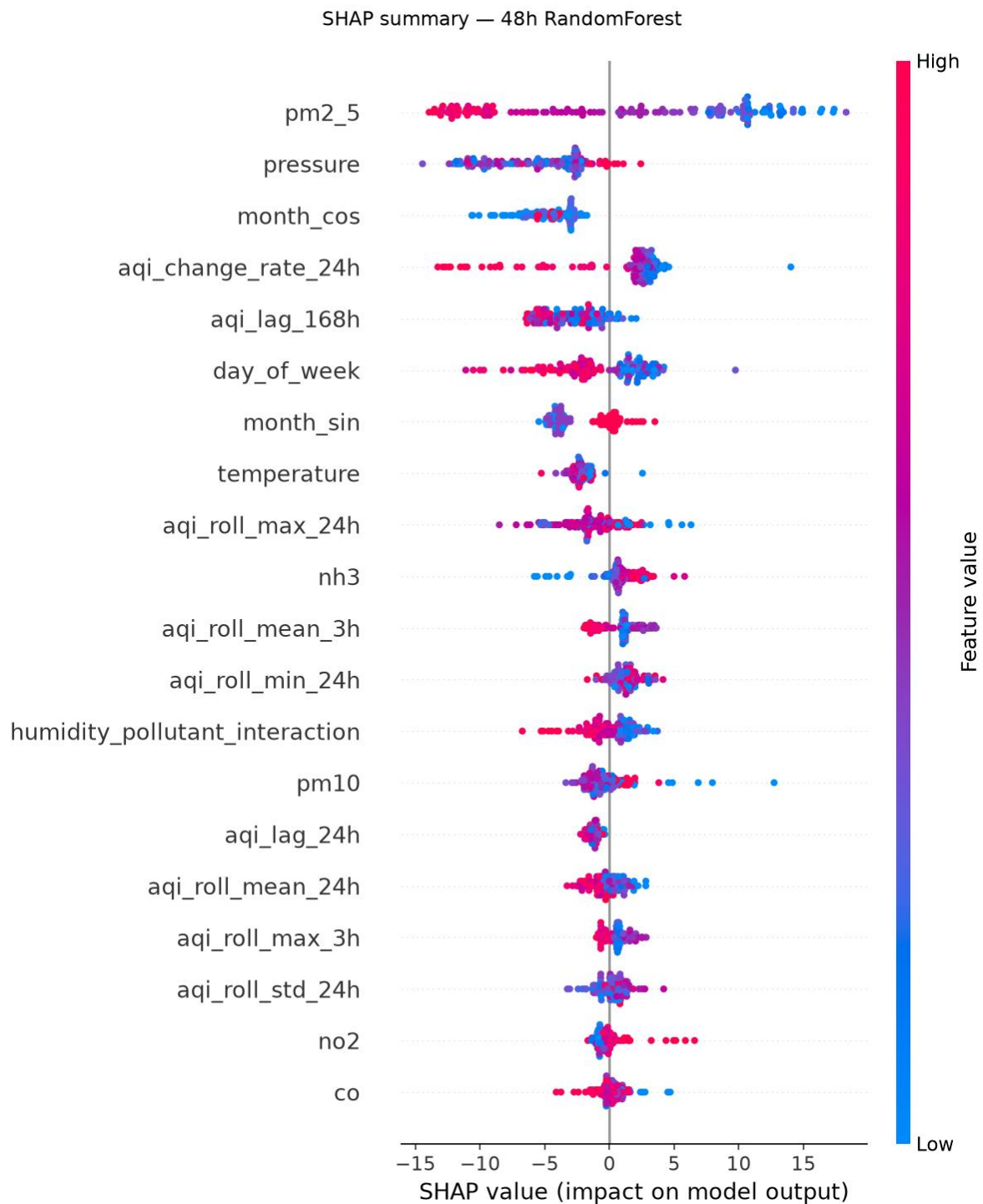


Figure 14: SHAP summary (beeswarm) for the 48h RandomForest model.

pm2_5 has the largest mean |SHAP| (8.68), then **pressure** (5.35) and **month_cos** (4.58). This ranking is for 150 sampled test rows.

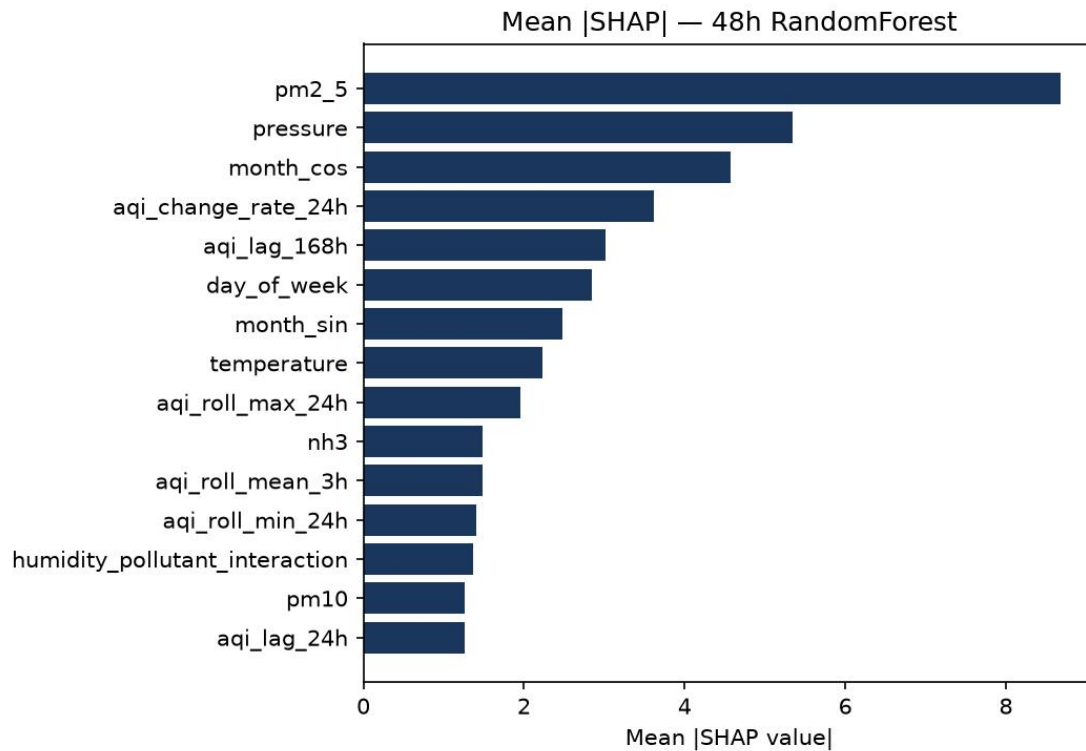


Figure 15: Mean |SHAP| bar chart for the 48h RandomForest model (top 15 features).

20.1.3 72-hour registered ensemble members

The 72h model is a mean ensemble. I ran TreeExplainer on each member because there is no single SHAP plot for the average.

Member	Top feature	Mean SHAP	SHAP rows
XGBoost_L1	month_cos	6.110957622528076	400
RandomForest	pm2_5	14.038097403347498	150
LightGBM	month_cos	5.241344214210389	400

Table 29: Highest mean |SHAP| feature per 72h ensemble member.

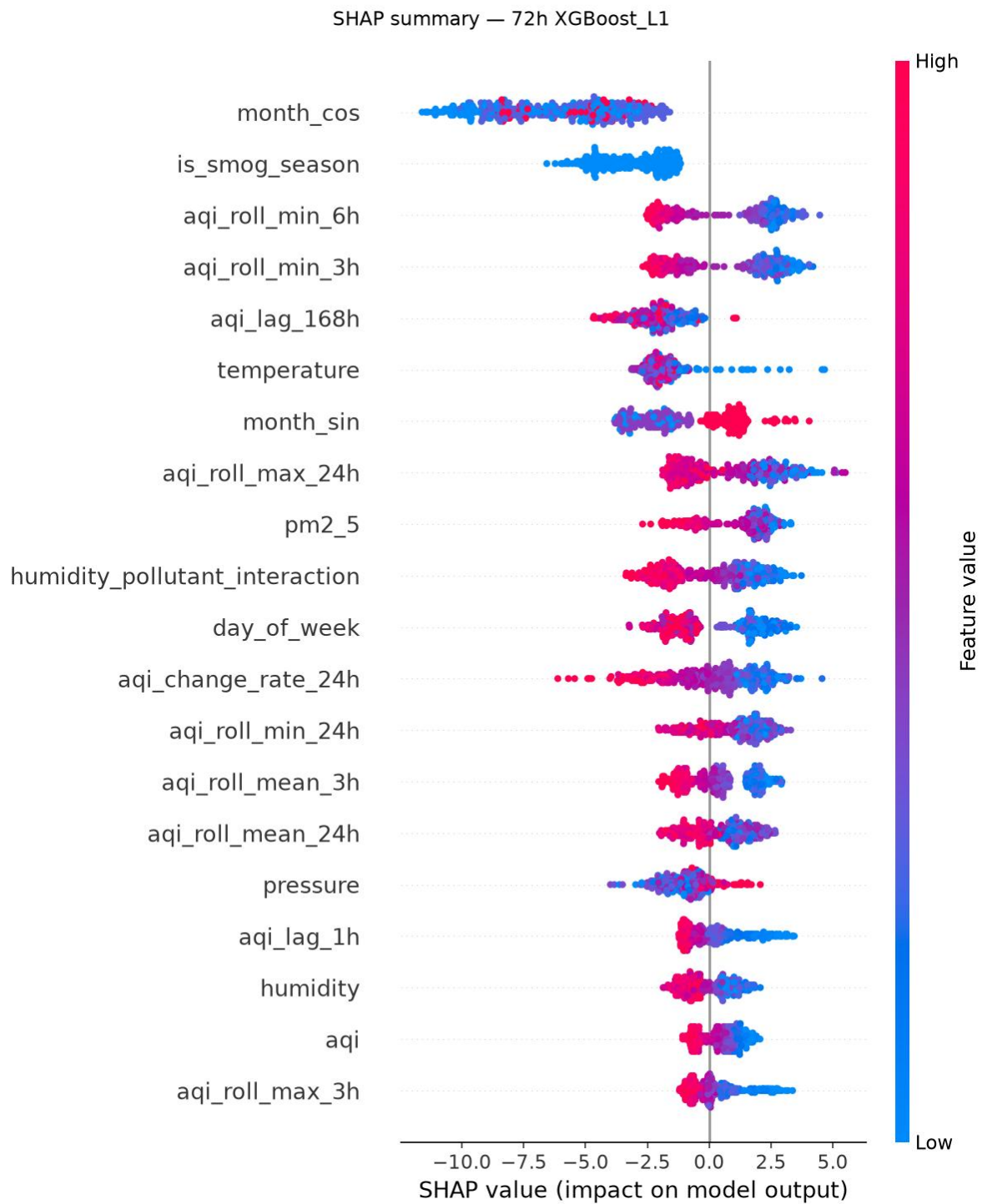


Figure 16: SHAP summary for 72h ensemble member XGBoost_L1.

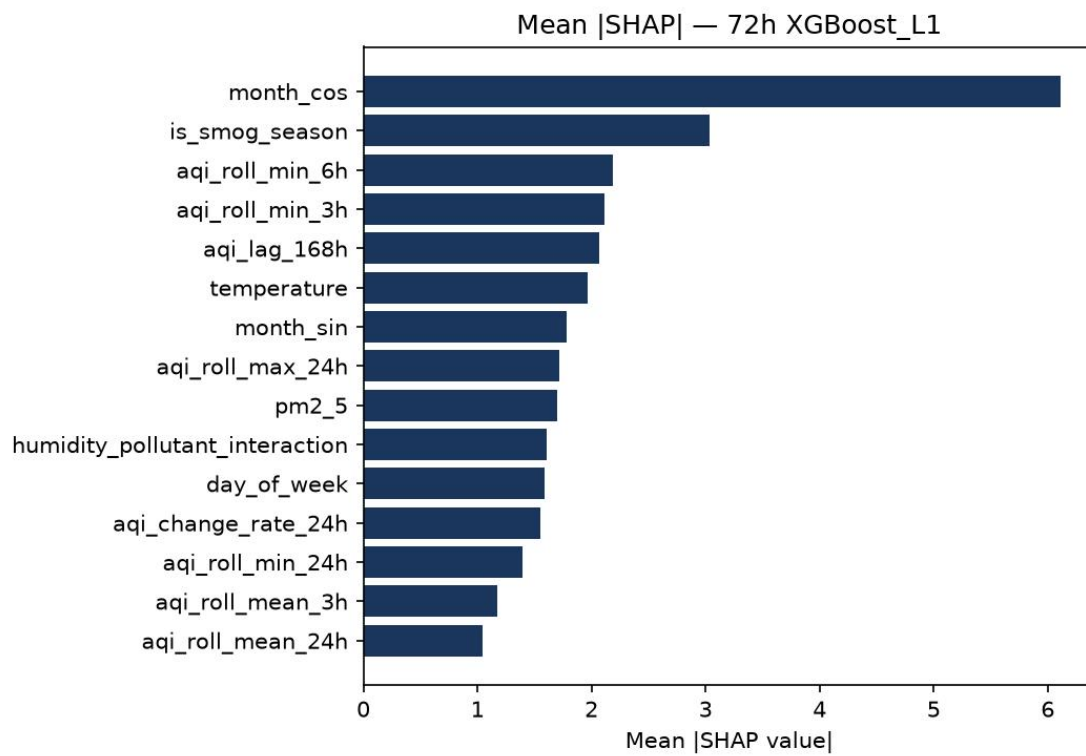


Figure 17: Mean |SHAP| bars for 72h XGBoost_L1.

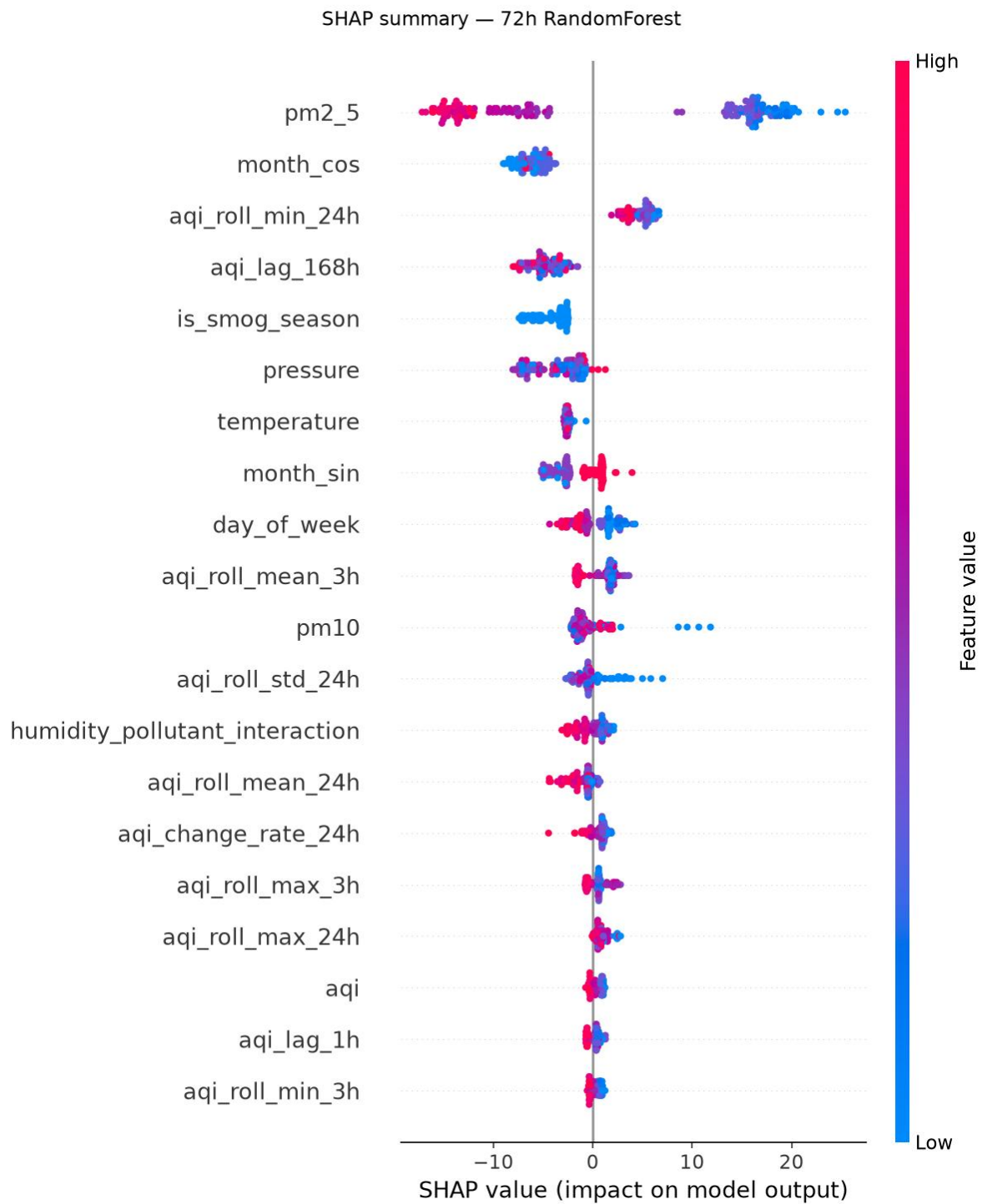


Figure 18: SHAP summary for 72h ensemble member RandomForest.

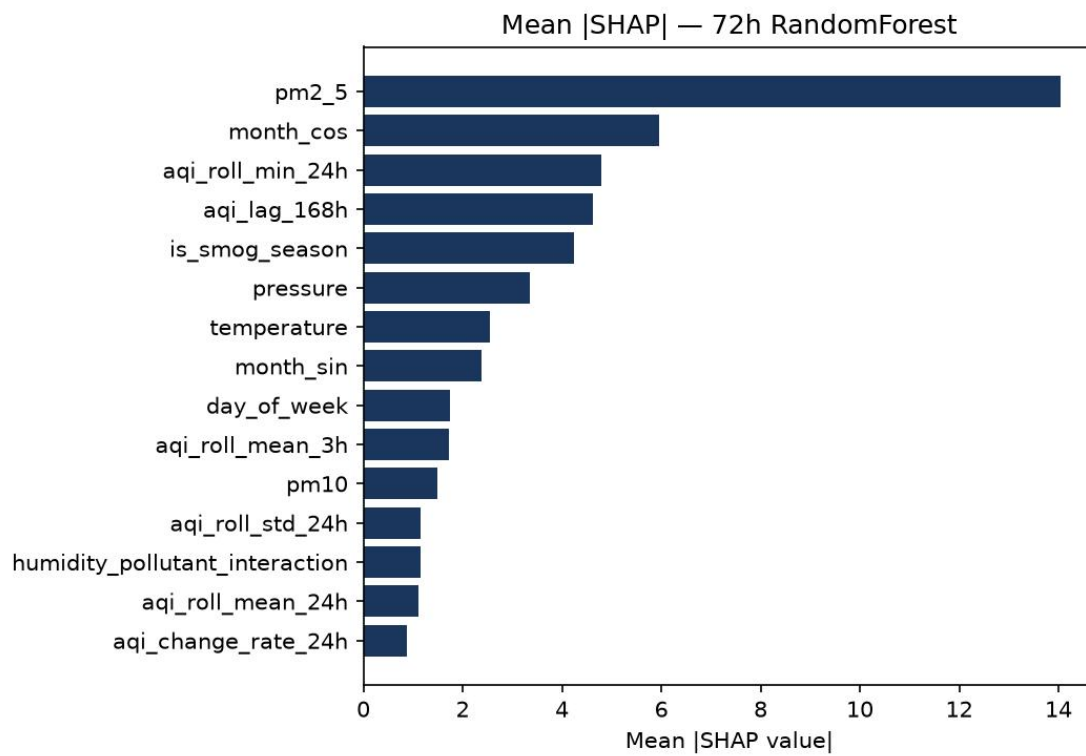


Figure 19: Mean |SHAP| bars for 72h RandomForest.

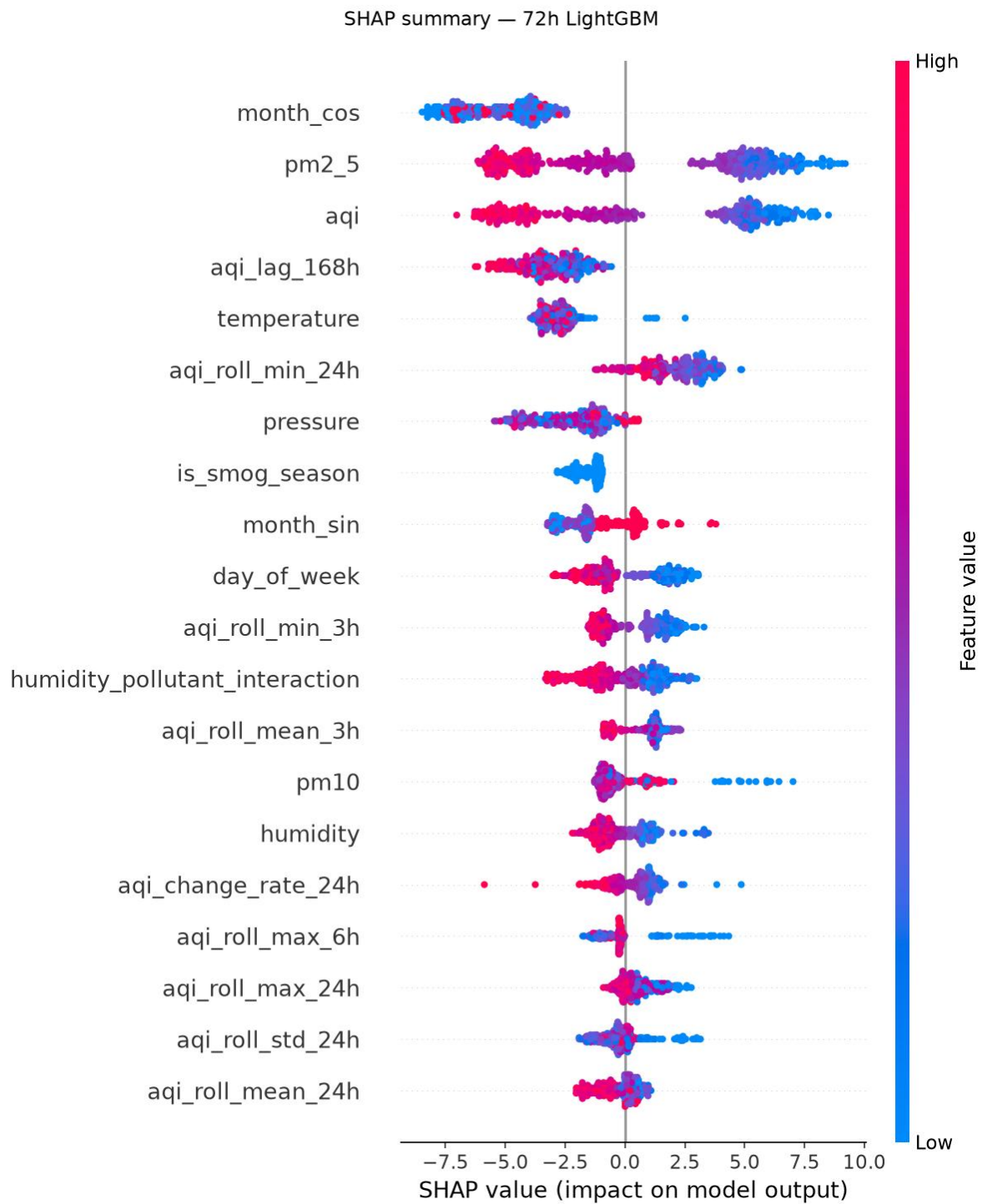


Figure 20: SHAP summary for 72h ensemble member LightGBM.

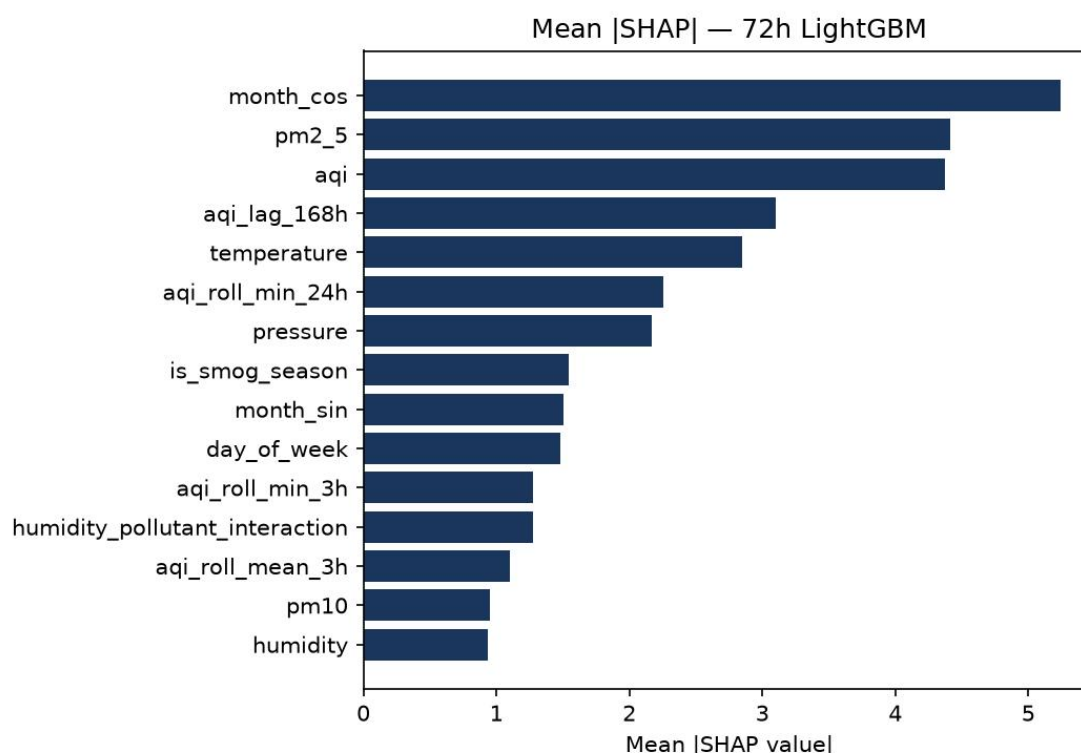


Figure 21: Mean |SHAP| bars for 72h LightGBM.

On these samples, **month_cos** is the top mean |SHAP| feature for XGBoost_L1 and LightGBM. **pm2_5** is the top feature for the RandomForest member (14.04). ``is_smog_season`` appears in the XGBoost_L1 and RandomForest top-10 lists. I am not reading these rankings as a causal story.

21. Classification Experiment

I also tried classification. The target was future OpenWeather class (openweather_aqi_category, values 1–5) at 24, 48 and 72 hours, using the same feature matrix as production. Persistence here means “tomorrow’s class equals today’s class.” I trained ordinal logistic regression, Random Forest, XGBoost, LightGBM, a majority-vote ensemble, LSTM and GRU with a 5-way softmax. A model counted as a winner only if it beat persistence on **accuracy, macro F1, RMSE, MAE and R²** together. RMSE, MAE and R² in this section are on the 1–5 labels, not on 0–500 AQI. Nothing from this experiment went into the registry.

21.1 Full-history classification

On the full archive (December 2020–August 2026, about 39,060 hours) persistence accuracy was **0.633 / 0.578 / 0.548** at 24 / 48 / 72 hours. Learned models sat around **0.31–0.38**. No model passed the five-metric gate. Persistence also has the lowest RMSE at every horizon (Table 30).

H	Model	RMSE	MAE	R ²
---	-------	------	-----	----------------

24	Persistence	0.729	0.420	0.412
24	Ordinal logistic	1.461	1.131	−1.358
24	Random Forest	1.307	0.964	−0.887
24	XGBoost	1.243	0.905	−0.707
24	LightGBM	1.322	0.972	−0.930
24	Ensemble	1.286	0.943	−0.827
24	LSTM	1.494	1.155	−1.462
24	GRU	1.472	1.127	−1.390
48	Persistence	0.816	0.500	0.264
48	Ordinal logistic	1.512	1.178	−1.525
48	Random Forest	1.493	1.150	−1.462
48	XGBoost	1.353	1.001	−1.022
48	LightGBM	1.334	0.977	−0.965
48	Ensemble	1.392	1.041	−1.141
48	LSTM	1.512	1.178	−1.524
48	GRU	1.503	1.161	−1.493
72	Persistence	0.863	0.546	0.178
72	Ordinal logistic	1.512	1.176	−1.525
72	Random Forest	1.510	1.173	−1.519
72	XGBoost	1.402	1.049	−1.171
72	LightGBM	1.397	1.050	−1.157
72	Ensemble	1.440	1.092	−1.291
72	LSTM	1.521	1.186	−1.552
72	GRU	1.513	1.173	−1.525

Table 30: Classification on full history. Scores are on OpenWeather classes 1–5. Persistence wins every horizon.

21.2 Post-break classification

On the April 2025–August 2026 window (about 9,866 hours) the race was closer. Only **72-hour Random Forest** passed all five metrics: accuracy **0.568** versus persistence **0.489**, RMSE **0.746** versus **0.895**. Forty-eight-hour GRU looked strong on accuracy (0.568 versus 0.486) but lost on macro F1 (0.292 versus 0.298). A walk-forward check (four folds, Random Forest versus persistence) passed the five-metric test in **1 of 12** fold×horizon slots, so the 72-hour win is not stable.

H	Model	RMSE	MAE	R ²
24	Persistence	0.826	0.520	−0.558
24	Ordinal logistic	0.753	0.472	−0.297
24	Random Forest	0.780	0.497	−0.391
24	XGBoost	0.821	0.539	−0.541
24	LightGBM	0.817	0.543	−0.527
24	Ensemble	0.813	0.532	−0.510
24	LSTM	0.778	0.495	−0.379
24	GRU	0.800	0.526	−0.455
48	Persistence	0.883	0.602	−0.789
48	Ordinal logistic	0.796	0.524	−0.456
48	Random Forest	0.812	0.534	−0.513
48	XGBoost	0.802	0.540	−0.476
48	LightGBM	0.829	0.560	−0.578
48	Ensemble	0.807	0.539	−0.496
48	LSTM	0.797	0.525	−0.488
48	GRU	0.758	0.480	−0.346
72	Persistence	0.895	0.607	−0.888
72	Ordinal logistic	0.776	0.502	−0.420
72	Random Forest *	0.746	0.474	−0.311
72	XGBoost	0.845	0.570	−0.683
72	LightGBM	0.868	0.594	−0.777

72	Ensemble	0.822	0.548	−0.592
72	LSTM	0.799	0.530	−0.495
72	GRU	0.767	0.489	−0.380

Table 31: Classification on the post-4 April 2025 window. *Only 72h Random Forest beat persistence on all five metrics (accuracy and macro F1 included).

Setting	Horizon	Persistence accuracy	Best learned (note)
Full history	24 h	0.633	Learned models ~0.31–0.38; none passed the gate
Full history	48 h	0.578	Learned models ~0.31–0.38; none passed the gate
Full history	72 h	0.548	Learned models ~0.31–0.38; none passed the gate
Post-break	24 h	—	No five-metric winner
Post-break	48 h	0.486	GRU accuracy 0.568, but macro F1 0.292 vs 0.298
Post-break	72 h	0.489	Random Forest accuracy 0.568 (passed all five; not stable in walk-forward)

Table 32: Classification accuracy summary. Macro F1 is stated where it decided the gate.

Classifying 1–5 is a different product from forecasting 0–500 AQI. It does not beat “today’s class” in a reliable way. I did not replace the continuous regression path.

22. Final Approach Selection

I compared regression and classification, each on the full archive and on the post-4 April 2025 window. Table 33 records what each experiment measured, which model won, and whether I kept it. The decision at the end of that comparison is the production system.

Experiment	What I measured	What won	Kept?
Regression, full history (first notebook)	RMSE, MAE, R^2 on 0–500 AQI	Persistence at 24 / 48 / 72 h (RMSE 66.73 / 84.80 / 92.46). Closest learned 24h model: XGBoost 74.36.	No
Regression, from 4 April 2025	RMSE, MAE, R^2 on 0–500 AQI	LightGBM 24h (RMSE 28.88), Random Forest 48h (30.54), top-3 ensemble 72h (31.65). All three beat persistence.	Yes
Classification, full history	Accuracy, macro F1, RMSE, MAE, R^2 on classes 1–5	Persistence (accuracy 0.633 / 0.578 / 0.548). Learned models ~0.31–0.38.	No
Classification, from 4 April 2025	Same five metrics on classes 1–5	Only 72h Random Forest passed the gate (accuracy 0.568 vs 0.489). Walk-forward: 1 of 12 slots.	No

Table 33: Experiments I ran and what I kept.

I selected **continuous US EPA AQI regression on the post-4 April 2025 window**, with **LightGBM at 24 hours**, **Random Forest at 48 hours**, and a **top-3 tabular ensemble at 72 hours**. Those are the models behind `aqi_forecaster_24h / 48h / 72h`.

Why this, and not the others:

- A forecast that loses to “no change” should not be served. Full-history regression lost to persistence. Classification mostly did too.
- RMSE and MAE are AQI points, which is what a user feels. Post-break regression error is about **29–32**, roughly half of full-history persistence error (**67–92**).
- Low R^2 on the post-break window is expected on a calm summer test set. RMSE/MAE still improve versus persistence, and that is what I used to choose.
- The 4 April 2025 break is visible in the raw series. Training through it asks the model to learn a volatility that no longer exists.

- OpenWeather 1–5 classification would change the product (five buckets instead of a 0–500 index) and did not beat class-persistence in a stable way.
- The registered models are tabular or ensemble payloads that the dashboard can serve from one feature row.

23. Why Other Models Were Not Selected

These reasons apply to the first notebook. The rule was the three-metric persistence gate.

23.1 Ridge

24/48/72 RMSE: 87.58 / 101.38 / 104.35 versus persistence 66.73 / 84.80 / 92.46. MAE and R^2 also fail at every horizon. Not registered.

23.2 RandomForest

24/48/72 RMSE: 84.91 / 118.05 / 116.12. All three metrics fail versus persistence at every horizon. Not registered.

23.3 XGBoost

24/48/72 RMSE: 74.36 / 97.21 / 111.99. Closest learned model at 24h and 48h, but RMSE and MAE remain higher than persistence and R^2 remains lower (24h R^2 0.684 versus 0.745). Not registered.

23.4 LightGBM

24/48/72 RMSE: 76.46 / 101.16 / 112.21. Fails the three-metric gate. Not registered.

23.5 Ensemble_top3_tabular

24/48/72 RMSE: 77.09 / 98.11 / 107.62. Fails the three-metric gate. Not registered.

23.6 LSTM

24/48/72 RMSE: 232.03 / 234.14 / 235.97 with negative R^2 . Not registered.

23.7 GRU

24/48/72 RMSE: 219.02 / 246.40 / 239.03 with negative R^2 . Not registered.

23.8 Prophet

24/48/72 RMSE: 2167.32 / 2159.44 / 2150.78. Fails by a large margin. Not registered.

24. Serving, Dashboard and Automation

Streamlit reads Hopsworks directly. FastAPI is optional. The dashboard is a single page. I run it locally with Streamlit, or the full stack from one Docker image (dashboard on port 8501,

API on port 8000). GitHub Actions refresh features hourly and retrain daily at 02:00 UTC. Feature-group version in those workflows is 4. The dashboard treats AQI **151** as the hazardous alert line.

The live app is <https://lahore-aqi-predictor.streamlit.app>.

25. Key Findings

- The raw snapshot has **46,419** rows and 16 stored columns from 27 November 2020 to 14 August 2026, with no duplicate timestamps and no missing values in stored columns.
- AQI is right-skewed (skewness 1.91), mean 247.8, median 177.
- pm2_5 correlation with AQI is **0.986**, which matches computing AQI from PM2.5.
- Smog-season (Oct–Jan) mean AQI is **399** versus **182** in other months.
- ADF on raw AQI rejects a unit root (statistic -11.76 , p-value 1.13×10^{-21}).
- The first notebook trained on aqi_features v3, 46,100 rows, 42 features, June-aligned split 27,947 / 8,075 / 10,078.
- Persistence test RMSE/MAE/R²: 24h **66.73 / 36.11 / 0.745**; 48h 84.80 / 45.78 / 0.588; 72h 92.46 / 51.52 / 0.510.
- **No learned model** beat persistence on RMSE, MAE and R² together. Nothing was registered from that run.
- Closest learned 24h model was XGBoost (RMSE 74.36).
- Prophet RMSE exceeded 2150 on every horizon. LSTM and GRU had negative R².
- OpenWeather breaks on 4 April 2025 (mean hourly |AQI change| $\sim 46 \rightarrow \sim 4.5$). I now start training there.
- On that later window, selected RMSE was 28.88 / 30.54 / 31.65. Those figures are not from the first notebook.
- I also ran classification of OpenWeather's 1–5 class on the same features. On full history, persistence accuracy was **0.633 / 0.578 / 0.548**; learned models sat around **0.31–0.38** and none passed the five-metric gate.
- On the post-break classification window, only **72h Random Forest** passed all five metrics (accuracy **0.568** versus **0.489**). Walk-forward held in **1 of 12** fold×horizon slots, so I did not adopt that path.
- The approach I selected is **continuous US EPA AQI regression from 4 April 2025**, with LightGBM (24h), Random Forest (48h) and a top-3 tabular ensemble (72h). Full-history regression and both classification windows were discarded because they did not beat persistence in a stable, serveable way.
- SHAP on the registered models: pressure (24h XGBoost), pm2_5 (48h RandomForest), month_cos / pm2_5 (72h ensemble members).

26. Limitations

- AQI is PM2.5-only, not the official EPA maximum across six criteria pollutants.
- Feature group v3 (the first training table) was built on a gappy frame, so about 13% of lags/targets spanned the wrong number of hours.

- About 3,700 missing hours in 415 outages; I interpolate at most 6 hours and drop longer gaps.
- The first notebook registered nothing because no model beat persistence on all three metrics.
- Prophet, LSTM and GRU cannot be served from a one-row predict payload.
- The production-window test months are a relatively calm warm-season stretch; R^2 at 48h and 72h on that window is slightly negative even for the selected models.
- The EDA notebook in git has no saved plots; I regenerated those figures from the parquet.
- SHAP for the first notebook does not exist (nothing registered). Figures 12–21 use 400 boosting rows and 150 RandomForest rows on the post-break split.
- Classification metrics (Section 21) are on OpenWeather classes 1–5, not on 0–500 AQI, and are not mixed with the regression tables.
- The 72h Random Forest classification win on the post-break window did not hold in walk-forward (1 of 12 slots), so I did not register a classifier.

27. Conclusion

I set out to forecast Lahore AQI at 24, 48 and 72 hours using OpenWeather pollution, Open-Meteo weather, Hopsworks storage, and a Streamlit dashboard with optional FastAPI. I compared two problem types — **continuous regression** of US EPA AQI from PM2.5 (0–500) and **classification** of OpenWeather’s 1–5 category — on both the full archive and the post-4 April 2025 window.

The raw series is right-skewed, tightly tied to PM2.5, worse in smog season, and strongly autocorrelated hour to hour. I engineered log pollutants, ACF-based lags, rolling windows, change rates, cyclical time features, a smog flag, weather interactions, and delta targets on an hourly grid.

On full-history regression (feature group v3, June-aligned split), **persistence had the best RMSE, MAE and R^2 at every horizon** (66.73 / 84.80 / 92.46). Closest learned 24h model was XGBoost (74.36). I registered nothing from that run.

On full-history classification, persistence accuracy was **0.633 / 0.578 / 0.548** and learned models sat around **0.31–0.38**. No classifier passed the five-metric gate. On the post-break classification window only **72h Random Forest** passed (accuracy 0.568 versus 0.489), and that win did not hold in walk-forward (1 of 12 slots).

I selected **continuous US EPA AQI regression from 4 April 2025 onwards**: LightGBM at 24 hours (RMSE **28.88**), Random Forest at 48 hours (**30.54**), and a top-3 tabular ensemble at 72 hours (**31.65**). I kept this path because those models beat persistence on AQI-point error, the regime break is visible in the raw series, 1–5 classification is a different product and was unstable, and the registered artefacts are serveable from one feature row. Full-history regression and both classification experiments remain in this report as discarded alternatives, not as competing production numbers.

SHAP on the models I actually serve is in Section 20.1. Pressure and 24-hour AQI change matter most for 24h XGBoost; PM2.5 for 48h RandomForest; the 72h ensemble members split between month_cos and PM2.5.